5

Urinary Tract Imaging

Basic Principles of Nuclear Medicine

Michael A. Gorin, and Steven P. Rowe

Keywords

dynamic renal imaging; functional imaging; molecular imaging; nuclear medicine; PET; positron emission tomography; single-photon emission computed tomography; single-photon imaging; SPECT

Questions

1. ^{99m}Tc-diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA) undergoes renal clearance by which mechanism?
    a. Glomerular filtration
    b. Active tubular secretion
    c. Anion exchange
    d. Both A and B
    e. All of the above
2. ^{99m}Tc-mercaptoacetyltriglycine (^{99m}Tc-MAG3) undergoes renal clearance by which mechanism?
    a. Glomerular filtration
    b. Active tubular secretion
    c. Anion exchange
    d. Both A and B

e. All of the above

3. ^{99m}Tc–dimercaptosuccinic acid (^{99m}Tc-DMSA) undergoes renal clearance and is endocytosed by cells of which part of the nephron?
   a. Glomerulus
   b. Proximal tubule
   c. Distal tubule
   d. Ascending loop of Henle
   e. Collecting duct
4. When performing ^{99m}Tc-MAG3 renography, the relative function of each kidney is determined by measuring the area under the time activity curve between what two time points post injection?
   a. 1 to 3 minutes
   b. 3 to 5 minutes
   c. 5 to 10 minutes
   d. 10 to 15 minutes
   e. 15 to 20 minutes
5. Which radiotracer is ideally suited for imaging renal scaring?
   a. ^{99m}Tc-DTPA
   b. ^{99m}Tc-MAG3
   c. ^{99m}Tc-DMSA
   d. ^{67}Ga-citrate
   e. ^{99m}Tc–sulfur colloid
6. 2-Deoxy-2-[^{18}F]fluoro-D-glucose (^{18}F-FDG) is phosphorylated by which glycolytic enzyme, trapping it inside metabolically active cells?
   a. Hexokinase
   b. Glucokinase
   c. Phosphofructokinase
   d. Pyruvate kinase
   e. Phosphoglucose isomerase
7. Which urologic malignancy is poorly imaged with ^{18}F-FDG positron emission tomography (PET)?
   a. Bladder cancer
   b. Penile cancer
   c. Prostate cancer

d. Renal cell carcinoma
e. Seminomatous germ cell tumor

8. What is the half-life of the radionuclide ^{18}F?
    a. 20.3 minutes
    b. 109.8 minutes
    c. 68.0 minutes
    d. 4.17 days
    e. 78.4 hours
9. Which of the following PET radiotracers for prostate cancer imaging is an amino acid analog?
    a. ^{18}F-DCFPyL
    b. ^{68}Ga-PSMA-11
    c. ^{11}C-choline
    d. ^{68}Ga-RM2
    e. ^{18}F-FACBC
10. Which protein is highly expressed by clear cell renal cell carcinoma (RCC) but not other renal tumor types and serves as a target for molecular imaging of this malignancy?
    a. GLUT1
    b. PSMA
    c. GRPR
    d. CAIX
    e. ASCT2
11. Which radiotracer has the highest specificity for imaging prostate cancer?
    a. ^{18}F-DCFPyL
    b. ^{67}Ga-citrate
    c. ^{11}C-choline
    d. ^{18}F-FDG
    e. ^{18}F-FACBC
12. ^{18}F-FDG PET is recommended for which application in men with testicular cancer?
    a. Initial cancer staging of patients with elevated postorchiectomy tumor markers

b. Initial cancer staging of patients with negative postorchiectomy tumor markers
c. Postchemotherapy imaging of a residual retroperitoneal mass in men with a seminomatous germ cell tumor
d. Postchemotherapy imaging of a residual retroperitoneal mass in men with a nonseminomatous germ cell tumors
e. Post-treatment surveillance in patients with negative markers

Answers

1. a. Goluerular filtration. ^{99m}Tc-DPTA is a single-photon radiotracer that is excreted in the urine and can be used to determine renal split function and to assess for functional obstruction. ^{99m}Tc-DTPA is extracted by the kidneys through glomerular filtration. The drug then quickly moves through the renal tubules and is excreted in the urine without being reabsorbed.
2. **b. Active tubular secretion.** ^{99m}Tc-MAG3 is a single-photon radiotracer that is excreted in the urine and can be used to determine renal split function and to assess for functional obstruction. ^{99m}Tc-MAG3 is protein bound in circulation and undergoes clearance exclusively through tubular secretion. ^{99m}Tc-MAG3 imaging is therefore not impacted by impaired glomerular filtration.
3. **b. Proximal tubule. ^{99m} Tc-DMSA is cleared from the plasma primarily by glomerular filtration.** Once within the lumina of the renal tubules, ^{99m}Tc-DMSA undergoes receptor-mediated endocytosis by the proximal tubular cells. The receptors responsible for the endocytosis of ^{99m}Tc-DMSA are megalin and cubilin.
4. **a. 1 to 3 minutes.** Dynamic renal imaging is performed in two phases. In the first phase, known as the perfusion phase, renal plasma flow to each individual renal unit is measured and compared with flow within the aorta. Activity should be

detected in the regions of interest overlying the kidneys within several seconds of detection in the aorta. The relative function of each kidney is determined by measuring the area under the time activity curves between 1 and 3 minutes post injection of the radiotracer.

5. **c. ^{99m}Tc-DMSA.** Because this radiotracer undergoes endocytosis by the proximal tubular cells, it is ideally suited for imaging cortical processes such as acute pyelonephritis and renal scaring.
6. a. Hexokinase. ^{18}F-FDG is a glucose analog that is taken up by metabolically active cells via GLUT transporters. Once within the cell, ^{18}F-FDG is phosphorylated by the glycolytic enzyme hexokinase, preventing diffusion back across the cell membrane. The trapped ^{18}F-FDG molecules, which are missing a 2-hydroxyl group, cannot undergo further glycolytic metabolism and remain intact allowing for their detection with PET imaging.
7. **c. Prostate cancer.** ^{18}F- FDG is taken up by malignant cells, which commonly shunt energy production toward anaerobic metabolism, a phenomenon referred to as the Warburg effect. A number of genitourinary malignancies, including urothelial carcinoma, renal cell carcinoma, squamous cell carcinoma of the penis, and testicular germ cell tumors, can be successfully imaged with ^{18}F-FDG PET. Prostate cancer is unique in that it does not show increased levels of glycolysis and therefore is poorly visualized with ^{18}F-FDG PET.
8. **b. 109.8 minutes.** The half-lives of other commonly used radionuclides for PET imaging can be found in Table 5.2.
9. **a. 18F-FACBC.** This radiotracer functions as a substrate for the amino acid transporters LAT1 and ASCT2, which are overexpressed by multiple malignancies including prostate cancer.
10. **d. CAIX. This cell surface protein is near universally expressed by clear cell RCC but not other renal tumor histologies.** CAIX is upregulated following loss of the von

Hippel-Lindau *(VHL)* gene, which is a defining event in the pathogenesis of clear cell RCC.

11. **a. 18F-DCFPyL.** PET radiotracers targeting prostate-specific membrane antigen (PSMA) have the highest specificity for prostate cancer imaging. Of the radiotracers listed, 18F-DCFPyL is the only PSMA-targeted radiotracer. A summary of other radiotracer used for prostate cancer imaging can be found in Table 5.3.
12. **c. Postchemotherapy imaging of a residual retroperitoneal mass in men with a seminomatous germ cell tumor.** Guidelines from the European Association of Urology endorse the use of ^{18}F-FDG PET for men with a residual mass following treatment of a seminoma with chemotherapy. Using ^{18}F-FDG PET in this manner helps to differentiate fibrosis from residual active tumor. For patients with nonseminomatous germ cell tumors, ^{18}F-FDG PET imaging is not indicated because there appears to be no clinical benefit for PET in the detection of viable tumor over the combination of computed tomography (CT) and serum markers.

Chapter Review

1. The most commonly used radiopharmaceutical agents for nuclear imaging of the kidneys are technetium ^{99m}Tc-diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA), ^{99m}Tc-mercaptoacetyltriglycine (MAG3), and ^{99m}Tc–dimercaptosuccinic acid (DMSA).
2. ^{99m}Tc-DTPA and ^{99m}Tc-MAG3 are used to measure renal blood flow, determine differential renal function, and to evaluate for the presence and degree of renal obstruction.
3. ^{99m}Tc-DTPA is cleared by glomerular filtration, whereas ^{99m}Tc-MAG3 is cleared by tubular secretion.
4. ^{99m}Tc-MAG3 is the preferred at most centers over ^{99m}Tc-DTPA because it has a higher extraction efficiency and is less impacted by changes in renal function

5. ^{99m}Tc-DMSA is retained by cells of the proximal renal tubules and is used to evaluate for infection and the presence renal scarring.
6. ^{99m}Tc-DTPA and ^{99m}Tc-MAG3 can also be used to evaluate renovascular hypertension, transplant graft function, and vesicoureteral reflux.
7. Molecular imaging of cancer is most commonly performed using the positron emission tomography (PET) radiotracer 2-deoxy-2-(^{18}F)fluoro-D-glucose (^{18}F-FDG)
8. A number of genitourinary malignancies can be successfully imaged with ^{18}F-FDG PET, albeit with varying degrees of clinical utility beyond conventional anatomical imaging techniques.
9. Because ^{18}F-FDG is excreted in the urine, imaging with this radiotracer is typically performed to detect distant sites of disease.
10. ^{18}F-FDG has little role in imaging prostate cancer and a number of other radiotracers have been developed for this purpose.
11. Radiotracers targeting PSMA are the most promising class of agents for prostate cancer imaging and in many parts of the world have become the new standard of care for imaging this malignancy.
12. One of the most well-established indications for ^{18}F-FDG PET imaging is in the detection of residual seminomatous germ cell tumors following chemotherapy.

6

Assessment of Urologic and Surgical Outcomes

David F. Penson, and Mark D. Tyson

Keywords

clinical epidemiology; health-related quality of life; outcomes research; patient-reported outcomes; study design

Questions

1. When assessing disease-specific mortality, researchers need to be aware that this outcome is subject to:
    a. detection bias.
    b. attribution bias.
    c. problems with loss to follow-up.
    d. all of the above.
    e. b and c.
2. Which of the following is NOT one of the four requirements for a valid surrogate endpoint as described by Prentice?
    a. Treatment is associated with the true endpoint.
    b. The surrogate endpoint is associated with the true endpoint.
    c. The full effect of treatment on the true endpoint is explained by the surrogate endpoint.
    d. The surrogate endpoint can reliably be measured using highly objective methods.
    e. Treatment is also associated with the surrogate endpoint.

3. With regard to patient-centered outcomes:
    a. clinicians can reliably assess these outcomes from the doctor–patient interaction.
    b. there are very few validated and reliable questionnaires available for use in urology.
    c. health-related quality of life is an important outcome to measure, but only function need be assessed to get a full portrait of this outcome.
    d. patient-centered questionnaires are developed using the principles of psychometric test theory.
    e. none of the above.
4. Which of the following methods has not been suggested as a technique to assess patient experience of pain?
    a. Collect verbal or written descriptions of the pain
    b. Query patient's caregiver regarding patient's pain level.
    c. Have subject rate pain via its effect on observable behavior.
    d. Use analogue techniques in which the patient compares his/her pain with an experimentally induced stimulus of a known intensity in a laboratory setting.
    e. All of the above can be used to assess pain levels.
5. When a scale has a coefficient α of 0.90, one can be assured that the scale has high degree of:
    a. alternate form reliability.
    b. test-retest reliability.
    c. internal consistency reliability.
    d. intraobserver reliability.
    e. interobserver reliability.
6. If a urologist were designing an instrument to measure cancer-specific quality of life following prostatectomy, how would she design an experiment to assess test-retest reliability?
    a. She could administer the questionnaire before and after the operation.
    b. She could administer the questionnaire to her patients and a group of age-matched healthy controls.
    c. She could use a differently worded questionnaire to obtain the same information.

d. She could administer the questionnaire to the same patient twice in the same week and compare the results.
e. She could administer the questionnaire to the same patient twice at an interval of one year and compare the results.

7. Of the following types of validity, which one often takes many years of instrument use by numerous investigators to adequately assess?
   a. Face validity
   b. Content validity
   c. Concurrent validity
   d. Predictive validity
   e. Construct validity
8. The use of uroflow results in validating a new survey instrument that measures disease-specific HRQOL in benign prostatic hyperplasia (BPH) patients is an example of:
   a. concurrent validity.
   b. content validity.
   c. construct validity.
   d. predictive validity.
   e. face validity.
9. A patient undergoes a radical cystectomy. On post-operative day #2, he aspirates and requires emergent intubation and transfer to the ICU. He is started on intravenous antibiotics and is extubated on post-operative day #4. The remainder of his hospital course in unremarkable and he is discharged on post-operative day #8. The complication he experienced is a Clavien-Dindo grade:
   a. II.
   b. IIIa.
   c. IIIb.
   d. IVa.
   e. IVb.
10. A patient with kidney cancer has a single target lesion, a 5-cm bony metastases. She is treated with a tyrosine kinase inhibitor (TKI). The lesion is reduced in size and is now 4 cm on imaging. According to the RECIST criteria, this is considered a:
    a. complete response.
    b. partial response.

c. stable disease.
d. non-CR/non-PD.
e. progressive disease.

Answers

1. **e. b and c.** When assessing disease-specific mortality, researchers need to be aware that this outcome is subject to attribution bias and problems with loss to follow-up.
2. **d. The surrogate endpoint can reliably be measured using highly objective methods.** It is not a requirement for a valid surrogate endpoint that it can reliably be measured using highly objective methods.
3. **d. Patient-centered questionnaires are developed using the principles of psychometric test theory.** For patient-centered outcomes, patient-centered questionnaires are developed using the principles of psychometric test theory.
4. **e. All of the above can be used to assess pain levels.** Collecting verbal or written descriptions of the pain; querying the patient's caregiver regarding the patient's pain level; having the subject rate pain via its effect on observable behavior; and using analogue techniques in which the patient compares his/her pain with an experimentally induced stimulus of a known intensity in a laboratory setting are all methods to assess patient pain.
5. **c. Internal consistency reliability.** When a scale has a coefficient alpha of 0.90, the scale has a high degree of internal consistency reliability.
6. **d. She could administer the questionnaire to the same patient twice in the same week and compare the results.** A urologist wishing to measure cancer-specific quality of life following prostatectomy could design an experiment to assess test-retest reliability by administering the questionnaire to the same patient twice in the same week and comparing the results.
7. **e. Construct validity.** Construct validity often takes many years of instrument use by numerous investigators to adequately assess.

8. **a. Concurrent validity.** The use of uroflow results in validating a new survey instrument that measures disease-specific HRQOL in BPH patients is an example of concurrent validity.
9. **d. IVa.** The patient's complication is Clavien-Dindo grade IVa.
10. **c. Stable disease.** This is considered stable disease per RECIST criteria.

Chapter Review

1. The effectiveness of health services delivery and treatment can be measured across three distinct dimensions: structure, process, and outcomes. Structure and process measures are easier to assess but outcomes tend to be most meaningful to clinicians and patients.
2. Mortality is the "hardest" endpoint one can assess in urology. That being said, it can be subject to bias. Specifically, studies using overall mortality can still be subject to lead- and length-time bias, and studies using disease-specific mortality may be subject to attribution bias.
3. While there are many proxy endpoints in urology, few meet all four requirements for being a valid surrogate endpoint. Despite this, urologists routinely use proxy endpoints in research and clinical practice.
4. There are a number of published and widely accepted criteria for defining disease progression and surgical complications in urology. While these urologists should use these reporting systems whenever possible, they should also remember that the use of these systems does not completely eliminate the potential for bias in research due to study design and other factors.
5. Frailty, functional status, and comorbidity are important potential confounders that should be considered in urologic research. There are numerous standardized tools available to capture these variables.
6. There are numerous patient-reported outcome tools available to assess symptoms and quality of life in patients with urologic diseases. Physicians and researchers should always use validated and reliable patient-centered tools when possible.

7

Ethics and Informed Consent

Vijaya M. Vemulakonda

Keywords

informed consent; medical ethics

Questions

1. The Hippocratic Oath does not include the ethical principle of:
    a. beneficence.
    b. non-maleficence.
    c. patient privacy.
    d. patient autonomy.
    e. accountability to the profession.
2. Important events leading to the modern framework of medical ethics include:
    a. The Nuremburg Trials.
    b. The Tuskegee Syphilis Studies.
    c. The Milgram Experiments.
    d. all of the above.
    e. none of the above.
3. The most important of the four basic principles of modern medical ethics is:
    a. justice.
    b. autonomy.
    c. beneficence.

d. non-maleficence.
e. none of the above.

4. The term bioethics was first coined by:
    a. Veatch.
    b. Potter.
    c. Beauchamp and Childress.
    d. Jonsen, Siegler, and Winslade.
    e. none of the above.
5. The Four Box Method does not:
    a. provide a systematic approach to apply ethical principles to specific clinical situations.
    b. include consideration of medical indications, patient preferences, quality of life, and other contextual features.
    c. create a venue for discussion of relative risks and benefits of potential treatment options.
    d. provide definitive answers to ethically challenging situations.
    e. none of the above
6. Clinical informed consent originated in:
    a. The Hippocratic Oath.
    b. 19th-century negligence laws.
    c. The Nuremburg Code.
    d. 20th-century battery laws.
7. Under current informed consent requirements, physicians are obligated to provide:
    a. information about the most common risks of a procedure.
    b. information that a "reasonable physician" would provide.
    c. information that a "reasonable person" would want to know.
    d. a written document detailing risks and benefits of the procedure.
8. The element of informed consent that is most often lacking is:
    a. description of the clinical problem, proposed treatment, and alternatives.
    b. discussion of the risks and benefits of the proposed treatment.
    c. assessment of the patient's understanding of the information provided by the medical provider.
    d. solicitation of the patient's preferences for treatment.

9. Exceptions to the informed consent requirement do not include:
    a. an unconscious 25-year-old patient who sustained multiple gunshot wounds.
    b. a 75-year-old patient with Alzheimer dementia who is being evaluated for benign prostatic hyperplasia (BPH) and has a previously identified surrogate decision maker.
    c. a 3-year-old patient who is accompanied by his parents for circumcision evaluation.
    d. a conscious 25-year-old patient who sustained multiple gunshot wounds.
    e. none of the above.
10. Issues thought to be associated with lack of patient understanding of consent include all EXCEPT:
    a. language of the surgical consent form is above the average patient reading level.
    b. patients may misunderstand the risks posed by surgery.
    c. patients idealize surgeons and defer to surgeon recommendations.
    d. patients may not recall the information contained within a surgical consent form.
    e. none of the above.
11. Use of decision aids have been associated with all of the following EXCEPT:
    a. improved knowledge of the procedure.
    b. improved assessment of risk.
    c. increased patient participation in the decision-making process.
    d. increased decisional conflict.
    e. none of the above.

Answers

1. **d. Patient autonomy.** Although the Hippocratic Oath includes the concepts of benefiting and avoiding harm to patients as well as protecting patient privacy, it does not include the concept of patient autonomy, which emerged in the 20th century.

2. **d. All of the above.** The Nuremburg trials revealed the involuntary experiments of Dr. Josef Mengele and other Nazi doctors on concentration camp detainees, resulting in pain, debilitation, and death of subjects. The trials led to the development of the Nuremburg code, which reinforced the need for voluntary consent, need to minimize harm, absence of other methods to obtain data, and need to terminate studies early if evidence of potential harm in a clinical research study. The Tuskegee studies were performed from 1932 to 1972 in Tuskegee, Alabama, to study the natural history of syphilis, with investigators withholding antibiotic treatment even after the discovery of the efficacy of penicillin to treat the disease. The study raised issues of adequate knowledge for consent, need to avoid coercion for participation, and need to terminate studies early if the investigator has knowledge of preventable harm. The Milgram experiments, which were conducted in 1963 at Yale by psychologist Stanley Milgram, encouraged subjects to administer fake electrical shocks to unseen participants. Recorded responses to the shocks were played, including screaming and silence suggestive of possible unconsciousness. Subjects were encouraged to continue to escalate shocks despite these responses, with the majority of subjects administering maximal shocks, leading to psychological trauma in many participants. The study raised concerns about use of deception in research and lack of adequate safeguards against harm. All of these events led to heightened scrutiny and development of heightened standards for medical research.
3. **e. None of the above.** According to Beauchamp and Childress, the four basic principles of autonomy, beneficence, non-maleficence, and justice should be considered equal in weight.
4. **b. Potter.** Van Rensselaer Potter first coined the term bioethics in 1970, described as the "study of the moral relationship between humans and their social and physical world."
5. **d. Provide definitive answers to ethically challenging situations.** Although the Four Box Method allows for systematic consideration of ethically complex clinical situations and creates a vehicle for discussions regarding treatment, it may also oversimplify ethical

problems and does not always provide clear “right” and “wrong” answers to address ethical concerns.

6. **d. 20th-century battery laws**. Although now considered within the scope of negligence, informed consent was first described by state courts in early 20th-century battery cases and was considered to be grounded in the right to bodily integrity. The Nuremburg code recommended expansion of the right to informed consent to medical research.
7. **c. Information that a “reasonable person” would want to know.** Although battery law required disclosure of risks and early negligence law required disclosure of what a “reasonable and prudent” physician would do, the current standard is what a reasonable person would want to know (Canterbury vs. Spence, 1972). The informed consent process requires both information from the physician and demonstrated understanding by the patient but does not require written documentation.
8. **c. Assessment of the patient’s understanding of the information provided by the medical provider.** Although all four elements are needed for a comprehensive informed consent, the majority of informed consents are incomplete with the element of assessing patient understanding of the information provided being the most commonly missed.
9. **d. A conscious 25-year-old patient who sustained multiple gunshot wounds.** In emergency settings where there is a significant threat to patient life or well-being, consent is presumed. However, this waiver of consent does not extend to the conscious patient even if refusal of treatment is life-threatening (In re Quackenbush 1978).
10. **e. None of the above.** All of the options have been posed as limitations to patient understanding of consent. The information provided in consent forms often is above the 12th-grade level, although the majority of patients have reading levels below the 8th-grade level. Additionally, patients who report reading consent forms may not recall the information contained in the document. Additionally, patients often underestimate their risks compared to other people or may use heuristics to make decisions that lead to a misunderstanding of the risks involved. Finally, patients often

accept surgical recommendations without meaningful participation in the decision-making process due to idealization of the surgeon and his or her judgment.

11. **d. Increased decisional conflict.** Decision aids have been associated with improved knowledge about procedures, reduced decisional conflict, improved patient participation in the decision, and improved assessment of risk.

Chapter Review

1. Although the scope of medical ethics continues to evolve, the basic ethical principles of autonomy, beneficence, non-maleficence, and justice remain the foundation of ethical medical practice.
2. Informed consent is essential to the practice of surgery with limited exceptions.
3. Despite the goal of ensuring adequate understanding of the nature, risks, benefits, and alternatives of surgery to facilitate a meaningful decision, the informed consent process is often incomplete. It is therefore the responsibility of the surgeon to make every effort to facilitate patient understanding and participation in the decision-making process.

PART II
Basics of Urologic Surgery

OUTLINE

8. Principles of Urologic Surgery: Perioperative Care
9. Principles of Urologic Surgery: Incisions and Access
10. Principles of Urologic Surgery: Intraoperative Technical Decisions
11. Lower Urinary Tract Catheterization
12. Fundamentals of Upper Urinary Tract Drainage
13. Principles of Urologic Endoscopy
14. Fundamentals of Laparoscopic and Robotic Urologic Surgery
15. Basic Energy Modalities in Urologic Surgery
16. Evaluation and Management of Hematuria
17. Complications of Urologic Surgery
18. Urologic Considerations in Pregnancy
19. Intraoperative Consultation

8

Principles of Urologic Surgery

Perioperative Care

Simpa S. Salami

Keywords

opioid abuse; patient safety; perioperative management; quality

Questions

1. A 64-year-old man is found to have an 8-cm left renal mass and presents to the office for evaluation regarding laparoscopic radical nephrectomy. He has a history of hypertension, non–insulin-dependent diabetes, and 30-pack-year history of tobacco use, which he quit 10 years ago. He has a strong family history of heart disease (father died at the age of 55 years from a myocardial infarction). Further questioning reveals that he does not regularly exercise but is able to walk up three flights of stairs without shortness of breath. Before surgery, to minimize the risk of complications, the patient should:
    a. undergo routine preoperative testing with complete blood count, basic metabolic panel, electrocardiogram, and chest radiograph.
    b. be referred to cardiology consultation to determine if further testing is necessary.

c. undergo noninvasive cardiac stress testing.
d. undergo pulmonary function testing to determine the need for preoperative bronchodilators.
e. be started on a perioperative β blocker to reduce the risk of perioperative myocardial ischemia.

2. With regard to unique patient populations, which of the following statements is TRUE?
   a. Although elderly patients have an increased perioperative risk, recent larger trials have not found age to be an independent risk factor for perioperative morbidity and mortality.
   b. Morbidly obese patients should undergo open rather than laparoscopic surgery because of increased risk of pulmonary complications.
   c. In a pregnant patient presenting with urolithiasis, operative intervention should be delayed, if possible, until the second trimester.
   d. A patient who presents with a 30-pound weight loss over the previous 3 months should be started on parenteral feedings immediately postoperatively after elective surgery.
   e. In patients with liver disease, the primary determinant of postoperative risk is degree of liver function enzyme abnormality.
3. A 74-year-old man with muscle-invasive bladder cancer is scheduled for radical cystectomy and ileal conduit urinary diversion. Preoperative urine culture shows no growth at 72 hours. The most important factor in the prevention of surgical site infection in this patient is:
   a. preoperative bowel preparation with oral antibiotics (Nichols prep) and sodium phosphate solution (Fleet).
   b. administration of 2 g cefoxitin 1 hour before incision.
   c. continuation of perioperative antibiotics for 48 hours following surgery.
   d. preoperative hair removal with mechanical clippers and proper sterile preparation of the operative field.
   e. optimization of comorbid illness and nutritional status.

4. According to current guidelines in the prevention of thromboembolic complications, a 78-year-old male with a recent history of colon cancer, medical history of hypertension, coronary artery disease (postoperative angioplasty with two coronary stents), and chronic renal insufficiency (creatinine, 2.9 mg/dL) undergoing laparoscopic transabdominal surgery should have pneumatic compression stockings and:
    a. early ambulation.
    b. aspirin and early ambulation.
    c. low-molecular-weight heparin.
    d. low-molecular-weight heparin and aspirin.
    e. unfractionated heparin and aspirin.
5. A 72-year-old woman with a history of asthma, mild congestive heart failure, and breast cancer is to undergo cystoscopy and placement of a midurethral sling. Of the following agents, the best choice for anesthesia induction would be:
    a. inhaled halothane.
    b. intravenous thiopental.
    c. inhaled desflurane.
    d. inhaled sevoflurane.
    e. intravenous succinylcholine.
6. A 56-year-old man with autoimmune disease is scheduled to undergo robotic-assisted laparoscopic radical prostatectomy. Which of the following represents the best choice for managing his chronic exogenous steroid therapy?
    a. Discontinue prior to surgery
    b. Discontinue prior to surgery and resume post-operatively
    c. Discontinue prior to surgery, administer stress dose steroid before the induction of anesthesia, and continue post-operatively until home medication is resumed
    d. Continue patient's oral medication at the current dosage through surgery
    e. Increase the dose of patient's oral medication prior to surgery and reduce back to baseline dose after surgery.

Answers

1. **e. Be started on preoperative β blocker to reduce the risk of perioperative myocardial ischemia.** This choice is best given the patient's multiple risk factors. Although cardiac stress testing may be considered, **the patient's ability to climb three flights of stairs indicates a capacity of greater than 4 metabolic equivalents (METs) and therefore a low risk of significant coronary artery disease.** Although routine preoperative testing is performed widely, there is no evidence that routine testing reduces the risk of perioperative complications.
2. **c. In a pregnant patient presenting with urolithiasis, operative intervention should be delayed, if possible, until the second trimester. The second trimester represents the least anesthetic risk to the mother and fetus with regard to spontaneous abortion and teratogenicity.** Although controversy exists as to the exact etiology, several recent trials have found age to be an independent predictor of morbidity on multivariate analyses. Laparoscopic surgery is the preferred approach in morbidly obese patients secondary to the reduced risk of pulmonary and wound complications. Literature suggests that **severely malnourished patients (>20 pounds weight loss in 3 months) significantly benefit from 7 to 10 days of enteral (not parenteral) feedings before elective surgery.** The primary determinants of the degree of severity in patients with cirrhosis are hepatic function and severity of clinical manifestations.
3. **b. Administration of 2 g cefoxitin 1 hour before incision.** Administration of appropriate antibiotics within 60 minutes of incision has been shown to significantly decrease the incidence of surgical site infections. Recent meta-analyses from the colorectal literature indicate that mechanical bowel preparation does not decrease the risk of postoperative infections. **Unless in the presence of active infection, perioperative antibiotics should be stopped 24 hours after incision** to decrease the risk of *Clostridium difficile* colitis. Although preoperative hair removal and

optimization of nutritional status and comorbid illness improve surgical outcomes, there is no specific evidence that this reduces surgical site infections.

4. **e. Unfractionated heparin and aspirin.** The clinical scenario describes a patient with high to highest risk of venous thromboembolism. Such a patient would require both mechanical and pharmacologic prophylaxis. **In a patient with renal insufficiency, unfractionated heparin is a better choice than low-molecular-weight heparin.** There is no evidence that aspirin is effective in the prevention of venous thromboembolism, but in a patient with coronary stents, aspirin is important in the prevention of stent thrombosis in the perioperative period.
5. **d. Inhaled sevoflurane.** This is an excellent choice for rapid induction in this patient secondary to its odorless and bronchodilation properties. Halothane can adversely affect left ventricular function and should be used with caution in patients with congestive heart failure. Desflurane has a pungent odor and is more suitable for maintenance of anesthesia during prolonged procedures. Intravenous thiopental can increase airway reactivity and is not appropriate in patients with asthma. Succinylcholine is appropriate for neuromuscular blockade and not commonly used for induction.
6. **c. Discontinue prior to surgery, administer stress dose steroid before the induction of anesthesia, and continue post-operatively until home medication is resumed.** Patients who have a depressed hypothalamic pituitary adrenal axis due to exogenous steroids should receive 50 to 100 mg of intravenous hydrocortisone before the induction of anesthesia and 25 to 50 mg every 8 hours thereafter until the patient's medication is resumed. Simply discontinuing the patient's steroid medication can result in adrenal crisis due to adrenal insufficiency post-operatively.

Chapter Review

1. One must always determine whether a woman in the childbearing years is pregnant before a surgical procedure. A urine pregnancy test is a simple method to do this.
2. The American Society of Anesthesiologists' classification is a significant predictor of operative mortality.
3. Preoperative cardiac evaluation is meant to identify serious coronary artery disease, heart failure, symptomatic arrhythmias, and the presence of a pacemaker or defibrillator. Major clinical predictors of cardiovascular risk are a recent myocardial infarction (within 1 month), unstable angina, evidence of an ischemic burden, decompensated heart failure, significant arrhythmias, and severe valvular disease.
4. A patient's ability to climb two flights of stairs is a good assessment of adequate functional capacity.
5. Patients with an FEV_1 of less than 30% predicted are at high risk for complications.
6. Smoking must be discontinued at least 8 weeks before surgery to be effective in reducing risk.
7. Perioperative β blockade is associated with a reduced risk of death among high-risk patients undergoing major noncardiac surgical procedures. However, more recent data bring this into question.
8. Patients who have a depressed hypothalamic pituitary adrenal axis due to exogenous steroids should receive 50 to 100 mg of intravenous hydrocortisone before the induction of anesthesia and 25 to 50 mg every 8 hours thereafter until the patient's medication is resumed.
9. In the elderly, delirium can be the first clinical sign of hypoxia or of metabolic or infectious complications.
10. In the pregnant patient, postoperative pain is best managed with narcotic analgesics.
11. A preoperative electrocardiogram should be obtained in all patients older than 40 years and in those with a significant cardiac history.
12. It is important to remember that for prophylaxis of venous thromboembolic disease and the use of antibiotic and mechanical bowel preps before intestinal surgery, the studies are often based on

data obtained from nonurologic patients. The urologist must consider this when the procedure being performed is significantly different from the standard general surgical operation on which the data are based. This is particularly true for bowel preparation, as urologic reconstructive procedures often require opening the isolated intestinal segment to be used in the procedure, exposing the entire contents to the operative field.

13. Parenteral antibiotics should be given within 60 minutes before intestinal surgery.
14. Nitrous oxide inhalation anesthesia results in bowel distention.
15. The half-life of warfarin is 36 to 42 hours, and it is recommended that warfarin be stopped 5 days before the surgical event.
16. Aspirin and clopidogrel are irreversible inhibitors of platelet function and should be discontinued 7 to 10 days before surgery if bleeding risk is to be minimized.
17. For moderate- to high-risk groups on anticoagulation therapy, a therapeutic bridge is performed using unfractionated or low-molecular-weight heparin. These may be stopped 4 or 12 hours, respectively, before the procedure and instituted shortly after its completion.
18. The indications for fresh frozen plasma are immediate reversal of warfarin and replacement of specific clotting factors.
19. The most common cause of transfusion-related fatality is transfusion-related acute lung injury (TRALI).
20. Hypothermia results in increased blood loss and an increased incidence of wound infection.
21. If hair is to be removed, it should preferably be removed immediately before the surgical event with mechanical clippers.
22. The need for postoperative parenteral nutrition should be anticipated in patients undergoing major urologic procedures involving the use of bowel. If it is likely the patient will not be able to take an adequate caloric intake orally by 7 to 10 days, postoperative parenteral nutrition should be instituted.
23. The second trimester represents the least anesthetic risk to the mother and fetus with regard to spontaneous abortion and teratogenicity.

24. Severely malnourished patients (>20 pounds weight loss in 3 months) significantly benefit from 7 to 10 days of enteral (not parenteral) feedings *before* elective surgery.
25. In a patient with renal insufficiency, unfractionated heparin is a better choice than low-molecular-weight heparin.

9

Principles of Urologic Surgery

Incisions and Access

David Mikhail, and Simon J. Hall

Keywords

abdominal wall incision; caval thrombectomy; Chevron Gibson; circumcision; flank incision; incision; inguinal incision; lumbodorsal lumbotomy; open anterior incision; open incision; open surgery; penile incision; perineal incision; retroperitoneal nephrectomy; scrotal incision

Questions

1. Which of the following is not considered an indication for an open abdominal approach (as opposed to minimally invasive)?
    a. Multiple prior abdominal surgeries
    b. Complex renal mass with caval thrombus
    c. Previous abdominal hernia repair with mesh
    d. Patient with multiple comorbidities
    e. Surgeon's preference and experience
2. Following a motor vehicle crash (MVC), a 35-year-old male is found to have a significant right-sided renal hilar injury on imaging. He becomes hemodynamically unstable, despite intravenous fluid resuscitation and massive transfusion protocol.

The decision is made to take him to the OR. What incision should you use?

a. Flank incision
b. Complete midline incision
c. Chevron incision
d. Subcostal incision
e. Thoracoabdominal incision

3. Which approach of abdominal wall fascial closure has been shown to have a higher rate of abdominal wall hernias?

a. Rapidly absorbable suture, running continuous closure
b. Rapidly absorbable suture, interrupted closure
c. Slowly absorbable suture, running continuous closure
d. Slowly absorbable suture, interrupted closure

4. A 25-year-old patient has been cleared to donate her left kidney to her cousin who recently was started on dialysis for end-stage renal disease. She is seeing you for pre-surgical consultation. You inform her you will be performing the nephrectomy laparoscopically and will be extracting the kidney through _____ because this has been shown to have _____.

a. extension of the inferior port site; lower morbidity and incisional hernia rates
b. extension of the midline port site; decreased pain scores and complications
c. pfannenstiel incision; lower morbidity and incisional hernia rates
d. pfannenstiel incision; decreased pain scores and complications
e. extension of the superior most port site; decreased pain scores and complications

5. A 45-year-old female had deceased donor renal transplant placed 5 years ago. She has been noncompliant with immunosuppressive medications and the allograft has failed. She has developed peri-allograft abscess, and the allograft must be removed. Which incision is best for renal allograft nephrectomy?

a. Lower midline
b. Inguinal incision
c. Subcostal incision

d. Gibson incision
e. Flank incision

6. Which incision is rarely used due to high risk of nerve injuries and ventral hernias?
   a. Midline incision
   b. Paramedian incision
   c. Thoraco-abdominal incision
   d. Pfannenstiel incision
   e. Gibson incision
7. Which of the following is NOT an advantage of the thoracoabdominal incision as an approach to large renal tumors?
   a. Exposure of adjacent thorax
   b. Exposure of retroperitoneum
   c. Early vascular control
   d. Large incision
   e. Access to inferior vena cava (IVC) for advanced disease/caval thrombus
8. Which is the second muscle layer incised in a typical flank incision?
   a. External oblique
   b. Internal oblique
   c. Serratus anterior
   d. Latissimus dorsi
   e. Transversalis
9. Which of the following is not considered a true flank incision approach?
   a. 12th rib supracostal
   b. 11th rib transcostal
   c. Subcostal
   d. 9th rib supracostal
10. Which nerve must be carefully handled to avoid injury during surgery in the inguinal canal?
   a. Femoral nerve
   b. Sciatic nerve
   c. Genitofemoral nerve
   d. Ilioinguinal nerve
   e. Lateral femoral cutaneous nerve

11. Which incision was used for the first-ever planned nephrectomy?
    a. Thoracoabdominal incision
    b. Flank incision
    c. Posterior lumbodorsal incision
    d. Subcostal incision
    e. Gibson incision
12. Which are NOT considered benefits of the dorsal lumbotomy approach compared to flank incisions?
    a. Rib and muscle sparing
    b. Less postoperative pain
    c. Decreased hospitalization
    d. Better surgical exposure for vascular control
    e. Decreased intra-peritoneal complications
13. Which of the following is NOT a border of the lumbodorsal region?
    a. 12th rib–superiorly
    b. Quadratus lumborum–inferiorly
    c. Spinal processes–medially
    d. Iliac crest–inferiorly
    e. Line between anterior superior iliac spine (ASIS) and costal margin–laterally
14. Which superficial muscles are NOT encountered during a dorsal lumbotomy approach?
    a. Internal oblique
    b. External oblique
    c. Latissimus dorsi
    d. Sacrospinalis
    e. Quadratus lumborom

Answers

1. **d. Patient with multiple comorbidities.** With an aging and more complex population, having multiple comorbidities on its own is not an indication for open surgery compared to minimally invasive surgery. Answers a, b, c, and e are all considered relative indications for open surgery. Patients with multiple previous

abdominal procedures are more likely to have adhesions and difficulties establishing a pneumoperitoneum. Complicated renal tumors with caval thrombi, although possible to do laparoscopically, should be considered for open surgery. A large abdominal wall mesh could significantly complicate a minimally invasive approach, and surgeon skill and preference is another important consideration for an open approach.

2. **b. Complete midline incision.** Trauma nephrectomies should always be approached with a laparotomy or complete midline incision. The other approaches would not be appropriate in this clinical setting.
3. **a. Rapidly absorbable suture, running continuous closure.** In a 2002 meta-analysis of closure techniques for midline abdominal incisions, Riet et al. found that continuous rapidly absorbable sutures had significantly more incisional hernias than slowly absorbable or non-absorbable sutures. No difference in hernia rates between slowly absorbable and non-absorbable sutures. (Reference: van ‘t Riet M, Steyerberg EW, Nellensteyn J, Bonjer HJ, Jeekel J: Meta-analysis of techniques for closure of midline abdominal incisions. *Br J Surg* 89:1350–1356, 2002.)
4. **c. Pfannenstiel incision; lower morbidity and incisional hernia rates.** A prospective study comparing Pfannenstiel incision versus port site expansion for nephrectomies showed that morbidity and length of stay were shorter in the pfannensteil group, while a meta-analysis for extractions in laparoscopic bowel surgery showed lower hernia rates with pfannensteil. (References: Binsaleh S, Madbouly K, Matsumoto ED, Kapoor A:· a prospective randomized study of pfannenstiel versus expanded port site incision for intact specimen extraction in laparoscopic radical nephrectomy. *J Endourol* 29[8]:913–918, 2015; Lee L, Abou-Khalil M, Liberman S, Boutros M, Fried GM, Feldman LS: Incidence of incisional hernia in the specimen extraction site for laparoscopic colorectal surgery: systematic review and meta-analysis. *Surg Endosc* 31[12]:5083–5093, 2017)

5. **d. Gibson incision.** Renal transplants and renal allograft nephrectomies are performed through a traditional Gibson incision, which gives extraperitoneal access to the iliac vessels and bladder.
6. **b. Paramedian incision.** Paramedian incisions have been described as an extra-peritoneal approach to nephroureterectomies and pediatric bladder augmentations. They are lateral to the rectus and are rarely used due to risks of nerve injury and ventral hernias.
7. **d. Large incision.** The thoracoabdominal incision, although considered to be a large invasive incision, provides the added benefit of significantly improved exposure, ability to achieve early vascular control, and access to major vessels (including the IVC) and organs for advanced renal tumors.
8. **b. Internal oblique.** The muscle and fascial layers encountered in a traditional flank incision, from skin to abdomen, are the external oblique, internal oblique, and transversalis muscles.
9. **d. 9th rib supracostal.** True flank incisions include the 12th rib supracostal, 11th rib transcostal, and subcostal approaches. Going above this level is often in the context of a thoraco-abdominal incision rather than a true flank incision.
10. **d. Ilioinguinal nerve.** The ilioinguinal nerve runs in the inguinal canal alongside the spermatic cord and should be identified upon opening and closure of an inguinal incision above the inguinal ligament. The genital branch of the genitofemoral nerve is present with the cord structures but is not exposed in the inguinal canal. The other nerves listed do not run through the inguinal canal.
11. **c. Posterior lumbodorsal incision.** The first-ever planned nephrectomy was performed in 1870 through a posterior lumbodorsal incision by Simon. This incision had multiple benefits over other open approaches. These include lack of muscle or rib distortion, faster convalescence, and decreased intra-peritoneal complications.
12. **d. Better surgical exposure for vascular control.** The dorsal lumbotomy approach, although not as common now, did boast multiple advantages over subcostal or anterior abdominal approaches. These include faster convalescence, less pain, less musculoskeletal complications (flank bulge), decreased

hospitalization, and less intra-peritoneal complications. The major disadvantage was limited surgical exposure to the renal hilum and vessels for vascular control.

13. **b. Quadratus lumborum–inferiorly.** The borders of the lumbodorsal region are the 12th rib superiorly, iliac crest inferiorly, spinous processes of vertebral columns medially, and a line between the ASIS (anterior superior iliac spine) and costal margin laterally. The incision is generally made directly over the quadratus lumborum, but it is not a border of this region.
14. **a. Internal oblique.** The superficial muscles encountered are the sacrospinalis (medially), latissimus dorsi (posteriorly), and external oblique (anteriorly). The incision is deepened through the lumbodorsal fascia where the sacrospinalis muscle is encountered and a "Y" should be made around it (Fig. 9.1). The incision is made directly over the quadratus lumborum muscle over the lateral part of the sacrospinalis muscle slightly obliquely toward the iliac crest or sacroiliac joint. The internal oblique muscle is not encountered during this incision.

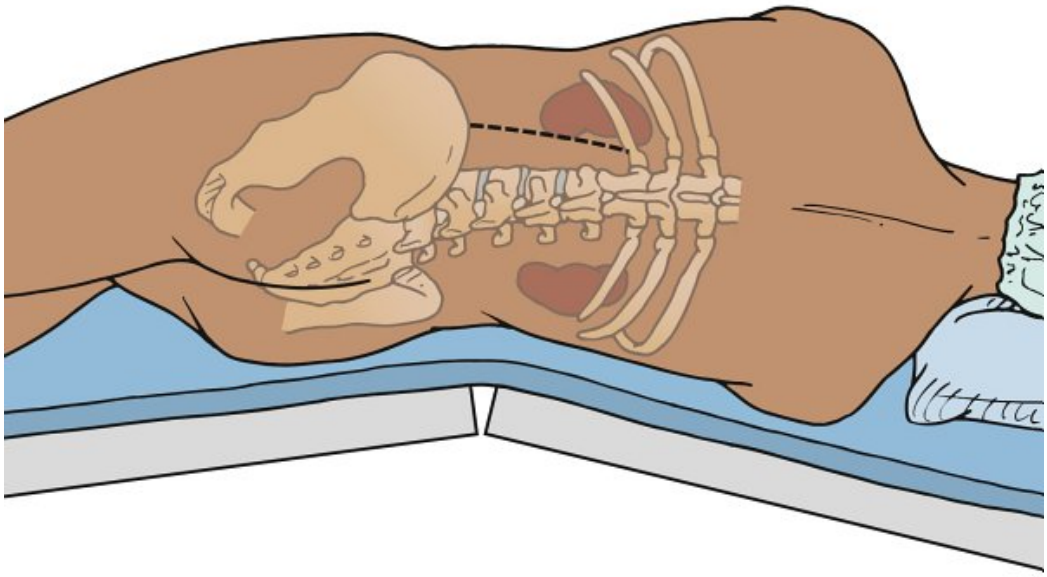


FIG. 9.1 Lumbodorsal incision.Once the iliac crest, the 12th rib, and the spinal processes have been located, the sacrospinalis muscle can be identified easily, and the surgeon can then locate the quadratus lumborum. This muscle goes from the medial part of the 12th rib to the middle third of the iliac crest; its upper third is under the sacrospinalis muscle. The skin incision is made over the quadratus lumborum muscle and begins at the costovertebral angle, over the lateral part of the sacrospinalis muscle, and with a slightly oblique course extends down to the iliac crest, 3 to 5 cm in front of the anterior margin of the sacrospinalis muscle.
Modified from Pansadora V: Surgical atlas: the posterior lumbotomy, surgery illustrated. *BJU Int* 95:1121–1131, 2005.

Chapter Review

1. There remain several indications for open surgical approaches in urology. Decisions regarding open versus minimally invasive approaches depend on complexity/size of pathology, patient preferences, and surgeon's skill set.
2. Knowledge of the abdominal wall musculature and anatomy are important to avoid key vessels and nerves when making incisions. These can usually be avoided by staying in the midline.
3. Anterior abdominal incisions can be used to access pelvic, intraperitoneal, retroperitoneal, and extraperitoneal structures.

Urologists most commonly use a lower midline incision for access to the bladder and prostate.

4. There are many open approaches to the kidney, each with their own advantages and disadvantages. Usually the kidney is approached via anterior or flank incisions, however dorsal incisions were a traditional approach which are being explored again by some groups.
5. Dorsal lumbotomy incisions traditionally have some advantages over anterior abdominal incisions, but with current minimally invasive options, this incision is used far less commonly. The lumbodorsal region is bordered between the 12th rib superiorly, iliac crest inferiorly, spinous processes medially, and an imaginary line between the ASIS and costal margin laterally.
6. Inguinal incisions can be above or below the inguinal ligament depending on the indication. They give access to contents of the spermatic cord, testicle, and ilio-inguinal lymph nodes.
7. Adult circumcisions are commonly performed through a two-incision technique on either side of the prepuce.
8. Meticulous hemostasis is necessary for scrotal and penile incisions as one can have significant bruising and bleeding complications.
9. Perineal incisions might become more common for various pelvic procedures with the development of new robotic technology.
10. A complete midline incision (laparotomy incision) is used for trauma nephrectomies. Gibson incisions are used for transplants and transplant nephrectomies.
11. When closing abdominal fascia, one should not use rapidly absorbable sutures as these have been linked to higher hernia rates.
12. Pfannenstiel incisions are better for specimen extractions than extension of a port site during minimally invasive procedures.
13. Thoracoabdominal incisions are usually made above the 9th to 10th rib bed, although they can be modified through the 8th rib bed and combined with a subcostal incision along with the standard longitudinal abdominal incision. A chest tube is not always required unless a parietal pleural injury is suspected.
14. Penile fractures can be approached through a degloving technique or by directly incising onto the suspected injury if the region is

confirmed with imaging.

15. With advancements in minimally invasive surgery and surgical technique, many incisions are being revisited for modern indications. These include lumbodorsal approach for partial nephrectomies and the perineal incision for robotic prostatectomies or cystectomies. Although the operative modality might have changed, all the incisions in this chapter should be known by urologists to have a plethora of options for any pathology.

10

Principles of Urologic Surgery

Intraoperative Technical Decisions

Manish A. Vira, and Christopher J. Hartman

Keywords

blood products; drains; hemostatic agents; intraoperative patient environment; suture material and incision closure

Questions

1. The most appropriate indication for blood product transfusion is:
    a. packed red blood cells for an 82-year-old male with coronary artery disease and hematocrit of 31%.
    b. fresh frozen plasma for a 69-year-old patient with an international normalized ratio (INR) of 1.6 scheduled to undergo laparotomy for a small bowel obstruction.
    c. platelet transfusion for a 78-year-old male with chronic renal insufficiency who was scheduled to undergo partial nephrectomy and found to have a platelet count of 55,000 on preoperative testing.
    d. packed red blood cells for a healthy 22-year-old male with a stable large retroperitoneal hematoma after motor vehicle accident and hematocrit of 21%.

e. Fresh frozen plasma for a 64-year-old female during resection of a large renal mass with inferior vena cava thrombus who experiences significant blood loss requiring 4 U of packed red blood cell transfusion.

2. To reduce the risk of iatrogenic injury to a patient in the operating room, the patient should:
    a. be maintained with core body temperature between 36°C and 38°C throughout the perioperative period.
    b. be instructed to bathe with an antiseptic solution the night before surgery.
    c. be secured to the operating room table with fixed shoulder braces for procedures in steep Trendelenburg.
    d. be positioned in the lithotomy position one leg at a time to ensure safe flexion of the hips.
    e. be positioned and draped by the operating room staff before arrival of the surgeon.
3. In a patient undergoing an exploratory laparotomy for pelvic abscess following radical cystectomy, the best method of abdominal fascial closure is with:
    a. polyglycolic acid (Dexon) suture with continuous closure.
    b. silk suture with continuous closure.
    c. polypropylene (Prolene) suture with interrupted closure.
    d. polyglactin (Vicryl) suture with interrupted closure.
    e. polydioxanone suture (PDS) with continuous closure.
4. A patient undergoing an open radical cystectomy with ileal conduit urinary diversion has a platelet count of 78,000/mm^3 prior to the procedure. During the surgery, diffuse oozing of blood is noted in the pelvis after the bladder is removed. The least appropriate hemostatic agent to use would be:
    a. An oxidized regenerated cellulose agent (Fibrillar, Surgicel, Nu-Knit).
    b. Microfibrillar collagen (Avitene).
    c. A topical thrombin agent (Thrombin-JMI, Evithrom, Recothrom).
    d. A fibrin sealant (Tisseel, Evicel, Evarrest).

e. A tissue adhesive (BioGlue).

5. A patient undergoing an open radical nephrectomy is noted to have a temperature of 35°C 3 hours into the procedure. Studies have demonstrated that this patient is at an increased risk of all of the following in the perioperative period except:
   a. increased bleeding.
   b. transfusion requirements.
   c. surgical site infection.
   d. prolonged hospital stay.
   e. renal insufficiency.
6. To avoid peripheral neuropathy during a robotic radical cystectomy in Trendelenburg position, the surgeon should make sure to:
   a. flex the patient's hips in lithotomy at no more than 70 degrees.
   b. abduct the patient's hips to between 45 and 55 degrees.
   c. pad the lateral knee to avoid excessive compression against the stirrups.
   d. tightly wrap the patient's arms at his or her side to prevent inadvertent crushing injuries from robotic arms.
   e. loosely secure the patient's shoulders to the table to avoid brachial plexus injury.

Answers

1. **c. Platelet transfusion for a 78-year-old male with chronic renal insufficiency who was scheduled to undergo partial nephrectomy and found to have a platelet count of 55,000 on preoperative testing.** This patient has moderate thrombocytopenia with likely platelet dysfunction secondary to uremia undergoing a high-bleeding-risk procedure; therefore platelet transfusion is indicated. Current indications for packed red blood cell transfusion are maintenance of hematocrit of greater than 30% in patients with high risk of myocardial ischemia or in patients with hematocrit 21% to 30% with signs of inadequate oxygen-carrying capacity. Fresh frozen plasma transfusion is indicated only in the presence of

active bleeding rather than isolated INR elevation or large-volume transfusion.

2. **a. Be maintained with core body temperature between 36°C and 38°C throughout the perioperative period.** Hypothermia by as little as 1°C has been shown to increase surgical site infection and postoperative complications. There is no evidence that showering with an antiseptic solution the night before surgery decreases the incidence of wound infection. Fixed shoulder braces have been associated with an increased risk of brachial plexus injury and should not be used in the operating room. Both legs should be positioned simultaneously when placing patients in the dorsal lithotomy position. Everyone in the operating room is responsible for patient safety, and therefore the surgeon should always be present for patient positioning.
3. **e. Polydioxanone suture (PDS) with continuous closure.** Continuous closure with PDS (slowly absorbable) suture has been shown to have the lowest wound failure rates. In the presence of infection, braided sutures (silk and Vicryl) should be avoided to prevent secondary wound infection and failure. Although nonabsorbable sutures may be used, these have been associated with increased postoperative patient discomfort. Fast-absorbing sutures (such as Dexon) should not be used in continuous fascial closure because of increased wound failure risks.
4. **b. Microfibrillar collagen (Avitene).** Microfibrillar collagen acts by enhancing platelet aggregation. Its efficacy is reduced in patients with thrombocytopenia and therefore may not stop bleeding in patients with thrombocytopenia as well as other hemostatic agents. Each of the other choices would be an acceptable hemostatic agent to use to in this case.
5. **e. Renal insufficiency.** Numerous studies have demonstrated that hypothermia may lead to complications and adverse effects in the perioperative period. Bleeding was found in one study to lead to an increased estimated blood loss of 16% and a 22% increase in transfusion requirements. In addition, hypothermia was associated with a threefold increased risk of wound infection and a 2.6-day increase in hospitalization. Renal insufficiency and other end-organ

impairment have not been demonstrated in patients with minor degrees of hypothermia in the perioperative period.

6. **c. Pad the lateral knee to avoid excessive compression against the stirrups.** Numerous injuries leading to neuropathy can occur during prolonged surgery in the Trendelenburg position. Of particular concern is injury to the brachial plexus from improper shoulder positioning, radial nerve injury from prolonged pressure on the radial nerve, and lower extremity nerve injuries from improper positioning in lithotomy stirrups. In general, the hips should be flexed at 80 to 100 degrees with 30- to 45-degree abduction. Tightly wrapping the patient's arms at his or her side imposes the risk of injury to the radial nerve. Only loosely securing the patient's shoulders to the table may prevent brachial plexus injuries, but it also puts the patient at risk of a fall off the table during the course of the procedure.

Chapter Review

1. The two primary reasons for hypothermia in the operating room are peripheral vasodilation that occurs with anesthetic agents and radiation of heat as the body interacts with the environment.
2. Normothermia is defined as a core temperature between 36°C and 38°C, and either hypothermia or hyperthermia may have adverse perioperative outcomes, including increased blood loss and increased incidence of surgical site infection.
3. Strategies to maintain normothermia include patient warming blankets, warmed intravenous and irrigation fluids, warmed CO_2 insufflation during laparoscopy, and an increase in ambient operating room temperature.
4. Commonly used skin antiseptic solutions for sterile skin preparation include alcohol, povidone-iodine, and chlorhexidine-based solutions, none of which has proven to be superior to the others in reducing SSI.
5. If hair is to be removed, it should preferably be removed immediately before the surgical procedure with clippers.

6. Three commonly occurring and preventable injuries in the operating room are retractor-associated injuries, thermal injuries, and patient position–related injuries.
7. Peripheral neuropathy may result from improper patient positioning and is often secondary to excessive stretch, prolonged compression, or ischemia of a nerve.
8. When positioning in lithotomy, care should be taken to manipulate both lower extremities simultaneously with flexion of the hips at 80 to 100 degrees with 30- to 45-degree abduction.
9. Current indications for packed red blood cell transfusion include hematocrit lower than 21%, maintenance of hematocrit of greater than 30% in patients with high risk of myocardial ischemia, or in patients with hematocrit 21% to 30% with signs of inadequate oxygen-carrying capacity.
10. Platelet transfusion may be beneficial in patients with severe thrombocytopenia ($<50{,}000/mm^3$) who are undergoing a planned surgical procedure, or in patients with moderate thrombocytopenia ($50{,}000/mm^3$ to $100{,}000/mm^3$) who are undergoing a high-risk procedure or have evidence of platelet dysfunction.
11. The indications for fresh frozen plasma are immediate reversal of warfarin, replacement of specific clotting factors, and evidence of bleeding in the setting of an INR greater than 1.5.
12. Adjunct hemostatic agents may be broadly classified as dry matrix agents or biologically active agents.
13. Oxidized regenerated cellulose agents (Fibrillar, Surgicel, Nu-Knit) aid in surgical hemostasis by activating the body's natural clotting cascade.
14. Topical thrombin agents (Thrombin-JMI, Evithrom, Recothrom) and the combination of thrombin sealants with a gelatin matrix (Floseal and Surgiflo) are not inhibited by urine and additionally resist stone formation within the urinary tract, making them advantageous hemostatic agents for use in urologic surgery.
15. Numerous studies have shown that surgical drains may not be of benefit in the perioperative period and may cause pain and lead to an

increased risk of SSI; therefore the routine use of surgical drains is not recommended outside of specific indications to do so.

16. The surgical scar regains 3% of strength at 1 week, 20% at 3 weeks, and 80% at 3 months.
17. Nonabsorbable sutures should be avoided when contact with urine is anticipated, such as during urinary tract anastomosis, because this increases the risk of stone development.
18. The risk of incisional hernia formation is not increased between slowly absorbable and nonabsorbable suture material in continuous versus interrupted fascial closures, although closure with nonabsorbable suture material was associated with increased wound pain.
19. Rapidly absorbable sutures used for continuous fascia closure are associated with an increased incidence of incisional hernias.
20. Fascia dehiscence generally occurs 7 to 10 days postoperatively but may occur up to 30 days after surgery.
21. In the presence of infection, braided sutures (Vicryl, Mersilene) should be avoided for wound closure.
22. Using staples for skin closure as opposed to sutures has been shown to decrease incision closure time, reduce the risk of wound infection, and has resulted in nonsignificant differences in cosmetic outcomes and patient satisfaction with wound appearance.

11

Lower Urinary Tract Catheterization

Joseph M. Jacob, and Chandru P. Sundaram

Keywords

catheter; catheterization; difficult catheter; difficult Foley; lower urinary tract; suprapubic catheter; urethral catheterization

Questions

1. The male urethra is ___ cm long.
    a. 15
    b. 13
    c. 24
    d. 20
2. The anterior male urethra is surrounded by ____
    a. the bulbospongiosus muscle.
    b. corpus spongiosum.
    c. both a and b.
    d. none of the above.
3. The female urethra is approximately ____ cm long
    a. 4
    b. 6
    c. 8
    d. 10
4. A 16-Fr catheter is approximately ____ cm in outer diameter.

a. 16
b. 10.4
c. 8.7
d. 5.3

5. _____ is the most rigid catheter material.
   a. Latex
   b. Silicone
   c. Polyvinylchloride (PVC)
   d. None of the above
6. Suprapubic catheter placement with a guide wire is the _____ technique.
   a. cystotomy
   b. Seldinger
   c. Lowsley
   d. trocar
7. The most common cause of inability to catheterize an Indiana pouch is _____.
   a. perforation
   b. catheter malfunction
   c. overdistension
   d. none of the above
8. A urostomy should never be catheterized.
   a. T
   b. F
9. Patients after bladder neck reconstruction should never be catheterized per urethra.
   a. T
   b. F
10. Catheter-associated urinary tract infection is the most common type of health care–associated infection
    a. T
    b. F
11. The incidence of bladder cancer in spinal cord patients with chronic catheters is _____.
    a. 1%
    b. 2%

c. 3%
d. 4%
e. 6%

Answers

1. **d. 20.** The male urethra is 18 to 20 cm. The anterior urethra is 16 cm.
2. **c. Both a and b.** The anterior male urethra is surrounded by both corpus spongiosum and bulbospongiosus muscle at the bulbar urethra.
3. **a. 4.** The female urethra is approximately 4 cm long
4. **d. 5.3.** To calculate French size approximately, divide by 3.
5. **a. Latex.** Rigidity from least to greatest: latex, silicone, PVC. This is important to understand, and different catheter materials may be more appropriate for different clinical situations.
6. **b. Seldinger.** The Seldinger technique involves gaining access into the bladder with a 21-gauge needle. I wire is placed once urine return is identified. Various catheters or kits can be used to place a catheter over the wire. The Lowsley retractor is a modified male urethral sound that can be used to place a suprapubic catheter. There are trocars that can be placed under vision with a cystoscope that can accommodate a catheter once the inner section is removed and the sheath left in place.
7. **c. Overdistension.** Overdistension is the most common reason for catheterization difficulty. If the catheter is forced, a perforation in the channel will be likely. A 21-gauge needle can be placed into the reservoir. After drainage, a catheter will usually pass without difficulty.
8. **b. F.** Catheterizing a urostomy gently can be performed without risk. Occasionally, catheterizing the stoma may be therapeutic in cases such as stomal stenosis or parastomal hernia.
9. **a. T.** A history of bladder neck reconstruction is one of the few contraindications to urethral catheterization.

10. **a. T.** Unfortunately, this is true based on a Centers for Disease Control and Prevention report. Decreasing catheter use should decrease the hospital-acquired infection rates.
11. **a. 1%.** There are retrospective studies that do show what appears to be an increased risk of bladder cancer. Some recommend surveillance after 8 years of chronic catheterization.

Chapter Review

1. There are many indications for urethral catheterization and many sizes and types of catheters that can be used.
2. As the French size increases, the diameter of the catheter lumen increases.
3. Catheter rigidity will vary by material.
4. Three-way hematuria catheters should be used if clot retention is suspect.
5. The urologist should be prepared to attain lower urinary drainage in almost all situations.
6. The clinician should closely review the patient's history and prior operative reports prior to instrumenting the lower urinary tract.
7. If benign prostatic hyperplasia (BPH) is expected, the practitioner should use an 18-Fr Coudé catheter for lower urinary tract drainage.
8. A history of bladder neck repair or closure are a contraindication to urethral catheterization.
9. Catheter placement should be avoided in patients with an artificial urinary sphincter (AUS). A 12-Fr catheter should be used after deactivation of the sphincter if necessary.
10. Catheter-associated urinary tract infections are the most common hospital-acquired infection.
11. Catheters may lead to trauma, stricture, and erosion.
12. Chronic catheterization can lead to bladder inflammation and increase the risk of bladder cancer.

12

Fundamentals of Upper Urinary Tract Drainage

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Keywords

nephrostomy tube; upper urinary tract drainage; ureteral stent

Questions

1. Percutaneous nephrostomy is not indicated for:
    a. instillation of intracavitary topical therapy for urothelial carcinoma.
    b. Whitaker test.
    c. management of fungal bezoars.
    d. urinary retention.
    e. ureteral injury.
2. Relative to retrograde ureteral stent placement, percutaneous nephrostomy:
    a. has a lower success rate.
    b. requires less anesthesia.
    c. is preferred in cases of ureteral obstruction owing to malignancy.
    d. is less commonly complicated by bacteriuria after indwelling for 1 week.
    e. is associated with worse health-related quality-of-life scores.

3. Which of the following is correct regarding the orientation of the kidney?
    a. The right kidney is slightly cephalad to the left kidney.
    b. The longitudinal axis is 45 degrees from vertical, with the lower pole lateral to the upper pole.
    c. The longitudinal axis is 45 degrees from vertical, with the lower pole anterior to the upper pole.
    d. The apposition of the colon to the kidney is greatest on the left side at the upper pole.
    e. Immediately posterior to the kidneys are the quadratus lumborum muscle, the psoas muscle, and the diaphragm.
4. Which of the following is correct regarding the intrarenal collecting system?
    a. Paired anterior and posterior calyces enter the infundibula approximately 90 degrees from each other.
    b. Compound calyces are most common in the lower pole
    c. Most kidneys have three distinct infundibula: the upper, middle, and lower.
    d. There are 8 to 16 minor calyces.
    e. There is a consistent relationship between anterior and posterior calyces and their medial-lateral position on anterior-posterior radiography.
5. The correct order of the division of the intrarenal branches of the renal artery is:
    a. segmental, arcuate, interlobar (infundibular), interlobular.
    b. segmental, arcuate, interlobular, interlobar (infundibular).
    c. segmental, interlobar (infundibular), arcuate, interlobular.
    d. interlobular, segmental, interlobar (infundibular), arcuate.
    e. segmental, interlobular, interlobar (infundibular), arcuate.
6. To reduce the risk of infectious complications from percutaneous renal surgery:
    a. all patients should receive prophylactic antimicrobials.
    b. urine cultures should be obtained on all patients.
    c. urine must be sterile before the procedure.
    d. gentamicin is an acceptable single agent for antimicrobial prophylaxis.

e. ampicillin/sulbactam is not an acceptable single agent for antimicrobial prophylaxis.

7. To reduce the risk of hemorrhagic complications associated with percutaneous renal access, the minimum recommended preoperative cessation period for:
   a. herbal medications is 2 weeks.
   b. clopidogrel is 10 days.
   c. aspirin is 5 days.
   d. warfarin is 5 days.
   e. nonsteroidal inflammatory agents is 1 day.
8. Which of the following have NOT been demonstrated in randomized controlled clinical trials to reduce pain associated with percutaneous renal access?
   a. Tract infiltration with local anesthetic
   b. Intercostal nerve block
   c. Thoracic paravertebral block
   d. Balloon dilation compared with semirigid plastic dilation of the access tract
   e. Smaller, compared with larger, caliber postprocedure nephrostomy tubes
9. An advantage of the supine versus prone position for percutaneous renal surgery is:
   a. improved pulmonary mechanics.
   b. a large horizontal working surface.
   c. easier entry into upper pole calyces.
   d. easier entry into posterior calyces.
   e. reduced pressure in the collecting system.
10. Access into which site provides the optimal versatility and safety for percutaneous renal surgery in the prone position?
    a. Upper pole posterior calyx
    b. Upper pole infundibulum
    c. Renal pelvis
    d. Middle calyx
    e. Lower pole anterior calyx
11. Techniques for retrograde assistance for percutaneous renal access include all but which of the following?

a. Straight ureteral catheter to inject air
b. Ureteral access sheath to facilitate drainage
c. Ureteroscopy to retrieve guidewire
d. Retrograde approach to percutaneous access
e. Retrograde placement of externalized (single pigtail) ureteral stent for drainage

12. Compared with an 18-gauge needle, the 21-gauge needle for percutaneous renal access:
    a. should not be used by inexperienced operators.
    b. requires a 0.025-inch guidewire.
    c. cannot be directed as easily.
    d. entails less risk of loss of access.
    e. is more traumatic.
13. Compared with ultrasonography, fluoroscopy for percutaneous renal access:
    a. is less suited in the morbidly obese.
    b. provides more rapid evaluation of the entire kidney.
    c. cannot be used to monitor tract dilation.
    d. visualizes the access needle better.
    e. is preferred in transplant kidneys.
14. The "triangulation" technique for fluoroscopic percutaneous renal access:
    a. increases radiation exposure to the operator's hands compared with the "eye-of-the-needle" technique.
    b. cannot be performed in malrotated kidneys.
    c. is not as dependent on retrograde assistance as the "eye-of-the-needle" technique.
    d. is less suitable than the "eye-of-the-needle" technique in morbidly obese patients.
    e. continuously monitors depth of needle penetration.
15. Dilation of the tract for percutaneous renal surgery is:
    a. not effective with a balloon dilator in hypermobile kidneys.
    b. most effective with semirigid dilator.
    c. least expensive with metal dilators.
    d. most rapid with metal dilators.
    e. easiest with a one-shot semirigid dilator.

16. When considering percutaneous renal surgery in horseshoe kidneys:
    a. upper pole access is dangerous.
    b. lower pole access is preferred in most cases.
    c. computed tomography can be misleading.
    d. the puncture site is more lateral than in normal kidneys.
    e. lower hemorrhage rates than in normal kidneys can be expected.
17. When considering percutaneous renal surgery in transplant kidneys:
    a. retrograde assistance is difficult.
    b. fluoroscopy is more useful than ultrasonography for initial access.
    c. the typical hypermobility renders tract dilation difficult.
    d. semirigid plastic dilators should not be used.
    e. secondary procedures are usually required.
18. Foley catheters for postprocedure nephrostomy drainage:
    a. do not need to be secured at the skin.
    b. can have a ureteral catheter passed through the end.
    c. should have the balloon filled with dilute contrast material.
    d. stay more securely in the kidney than Malecot catheters.
    e. are less likely to become infected than Malecot catheters.
19. The Cope retention mechanism:
    a. is used in nephroureteral stents.
    b. is used in internal ureteral stents.
    c. is more secure than a balloon catheter.
    d. requires cutting the tube to disengage.
    e. should not be used in more than one access site.
20. Alternatives to a nephrostomy tube after percutaneous renal surgery include all EXCEPT:
    a. maintenance of the working sheath.
    b. an internal ureteral stent that is removed cystoscopically.
    c. an internal ureteral stent with an attached string that exits out the flank.
    d. a ureteral stent externalized out the urethra.
    e. no drainage tube at all.
21. A postoperative nephrostomy tube:

a. offers greater assurance of upper urinary tract drainage than an internal ureteral stent.
b. should be placed in the dilated access site.
c. does not maintain the percutaneous access tract unless >18 Fr.
d. reduces postoperative bleeding.
e. is associated with pain unrelated to tube diameter.

22. A small-caliber (8 to 18 Fr) compared with a large-caliber (20 to 24 Fr) nephrostomy tube after percutaneous renal surgery is associated with:
    a. equivalent pain.
    b. more urinary leakage.
    c. less postprocedure blood loss.
    d. less need for removal in the radiology suite.
    e. earlier hospital discharge.
23. Adjuncts intended to enhance hemostasis of the percutaneous tract include all EXCEPT:
    a. direct cauterization of the tract.
    b. microwave treatment of the tract.
    c. cryotreatment of the tract.
    d. insertion of oxidized cellulose.
    e. instillation of fibrin glue.
24. Compared with internal ureteral stents after percutaneous renal surgery, nephrostomy tubes are associated with:
    a. reduced need for a second procedure for removal.
    b. greater technical success rate.
    c. greater narcotic use.
    d. fewer complications.
    e. less urinary leakage from skin entry site.
25. Following an unremarkable percutaneous nephrolithotomy, there is nonpulsatile bleeding from the tract when the sheath is removed around a 12-Fr nephrostomy tube. The next step is:
    a. replace the nephrostomy tube with an 18-Fr Malecot catheter.
    b. replace the nephrostomy tube with a ureteral stent and suture the skin.
    c. irrigate the nephrostomy tube.

d. occlude the nephrostomy tube and apply pressure to the incision.
e. replace the nephrostomy tube with a Kaye nephrostomy tamponade balloon.

26. During a percutaneous resection of a 2-cm upper pole urothelial neoplasm, there is sudden hemorrhage from the resection site. The next step is:
    a. continue with the procedure if vision is adequate.
    b. insert a percutaneous nephroureteral stent.
    c. instill gelatin granules plus thrombin into the collecting system.
    d. place an 18-Fr Councill catheter with the balloon inflated at the injury site.
    e. prepare the patient for selective angioembolization.
27. A 65-year-old man calls the office 1 week after percutaneous nephrolithotomy complaining of bright red blood in the urine on his last two urinations. He is otherwise feeling well. He should next:
    a. check the percutaneous access site and come to the hospital if there is external bleeding.
    b. force fluids and call back if bleeding persists.
    c. take aminocaproic acid (Amicar).
    d. apply pressure to the percutaneous access site.
    e. come to the hospital.
28. Which of the following has NOT been reported to cause renal pelvic perforation in association with percutaneous renal surgery?
    a. Wire passage
    b. Tract dilation
    c. Massive hemorrhage
    d. Use of resectoscope
    e. Ultrasonic lithotripsy
29. Two days after percutaneous endopyelotomy in a 65-year-old woman, nephrostography reveals contrast entering the colon. The next step is to:
    a. perform exploratory laparotomy.

b. maintain the nephrostomy tube in place and insert a ureteral stent.
c. maintain the nephrostomy tube in place and insert a colostomy tube.
d. back out the nephrostomy tube into the colon and insert a new nephrostomy tube.
e. start parenteral feeding, after appropriate tube insertions.

30. Injury to which organ during percutaneous renal surgery can often be managed with little additional interventions?
    a. Liver
    b. Spleen
    c. Duodenum
    d. Jejunum
    e. Gallbladder
31. Regarding pleural injuries in association with percutaneous renal surgery:
    a. access below the 12th rib results in hydropneumothorax in 1% to 2% of cases.
    b. supra-12th rib punctures (the 11th intercostal space) result in hydropneumothorax in 20% to 40% of cases.
    c. supra-11th rib punctures (the 10th intercostal space) result in hydropneumothorax in 50% to 75% of cases.
    d. combined with distal ureteral obstruction, a nephropleural fistula can occur.
    e. thoracostomy to water seal drainage and suction is recommended.
32. Irrigation fluid during percutaneous renal surgery:
    a. is not absorbed systemically unless there is significant venous injury.
    b. should be normal saline except during percutaneous nephrolithotomy.
    c. can have fatal consequences.
    d. should not be glycine.
    e. will not create a defined extrarenal collection.
33. A 55-year-old woman has an oral temperature of 38.5°C on the first night after an uncomplicated percutaneous nephrolithotomy for a

partial staghorn renal calculus. A nephrostomy tube is in place. She is hemodynamically stable. The preoperative urine culture had grown a pansensitive *Proteus* sp., and she had received oral trimethoprim sulfamethoxazole for 2 weeks preoperatively. One gram of cefazolin had been administered on call to the operating room. The next step is:

a. observation.
b. culture aspirate from nephrostomy tube and irrigate nephrostomy tube.
c. Doppler ultrasonography of lower extremities and/or pulmonary embolus-protocol computed tomography scan.
d. administer broad-spectrum antibiotics.
e. culture urine and blood, obtain chest radiograph, and administer broad-spectrum antibiotics.

34. Following percutaneous renal surgery, loss of renal function is:
    a. approximately 5% of ipsilateral function per access site.
    b. minimal in the absence of vascular injury.
    c. greater than after shock wave lithotripsy.
    d. less in pelvic compared with orthotopic kidneys.
    e. greater in solitary compared with nonsolitary kidneys.
35. With regard to ureteral stents, which factors have been associated with improved patient tolerance?
    a. Ureteral stent length
    b. Ureteral stent diameter
    c. Stent material
    d. Distal stent tip architecture (pigtail vs. Polaris)
    e. None of the above

Answers

1. **d. Urinary retention.** Obstruction of the lower urinary tract is best treated by drainage of the bladder rather than the kidney, unless secondary obstruction of the upper tract has developed that is refractory to vesical drainage. The other indications are appropriate ones for percutaneous nephrostomy.

2. **b. Requires less anesthesia.** Percutaneous nephrostomy can be done under local anesthesia, as opposed to retrograde ureteral stent placement, which usually requires at least intravenous sedation, and commonly general or regional anesthesia. Percutaneous nephrostomy has a greater initial success rate than retrograde ureteral stent placement, at least when the collecting system is dilated. Percutaneous nephrostomy is commonly associated with bacteriuria and has health-related quality-of-life scores that are equivalent to those associated with retrograde ureteral stent placement. Ureteral stents provide satisfactory drainage in most cases of ureteral obstruction owing to malignancy.
3. **e. Immediately posterior to the kidneys are the quadratus lumborum muscle, the psoas muscle, and the diaphragm.** The upper poles are anterior to attachments of the diaphragm. It is the left kidney that is slightly cephalad to the right one. The second two statements are correct, except that the angulation is 30 degrees rather than 45 degrees. The apposition of the colon to the kidney varies with location; it is greatest on the left side but at the lower rather than upper pole.
4. **a. Paired anterior and posterior calyces enter the infundibula approximately 90 degrees from each other.** The paired anterior and posterior calyces enter approximately 90 degrees from each other. **Although compound calyces are common in the lower pole, they are almost always present in the upper pole.** In approximately two-thirds of kidneys, there are only two major calyceal systems (upper and lower). There are 5 to 14 minor calyces in each kidney. Because variation is considerable, the lateral-medial orientation of the calyces on anteroposterior radiography cannot be used to reliably determine which calyces are posterior.
5. **c. Segmental, interlobar (infundibular), arcuate, interlobular.**
6. **a. All patients should receive prophylactic antimicrobials.** The American Urological Association recommends periprocedural antimicrobial prophylaxis for all cases of percutaneous renal surgery. Urine cultures are considered standard only in patients where bacteriuria is likely; in other cases a screening urinalysis

likely is adequate, with urine culture when the urinalysis is suspicious. The urine cannot be sterilized in some patients, especially in the presence of an externalized urinary catheter or an infected calculus, and the goal in these situations is only to suppress the bacterial count before intervention. Aminoglycosides (e.g., gentamicin) are acceptable for antimicrobial prophylaxis when combined with another agent. Ampicillin/sulbactam, first- and second-generation cephalosporins, and fluoroquinolones are acceptable single agents for antimicrobial prophylaxis.

7. **d. Warfarin is 5 days. The recommended preoperative cessation periods are as follows: herbal medicines, 1 week; clopidogrel, 5 days; aspirin, 1 week; warfarin, 5 days; nonsteroidal inflammatory agents, 3 to 7 days.**
8. **d. Balloon dilation compared with semirigid plastic dilation of the access tract.** There is no evidence that balloon dilation is associated with less pain compared with semirigid plastic dilation of the access tract. All of the other maneuvers have been demonstrated in randomized controlled clinical trials to reduce pain associated with percutaneous renal access.
9. **e. Reduced pressure in the collecting system.** The angle of the sheath is more horizontal in the supine compared with the prone position for percutaneous renal surgery, which reduces pressure in the collecting system (the volume also is reduced, which is a disadvantage). When padding is appropriate, pulmonary mechanics are better in the prone position. The prone position also provides a large horizontal working surface and easier entry into posterior and upper pole calyces compared to the supine position.
10. **a. Upper pole posterior calyx.** This offers the most versatile access to the intrarenal collecting system, and as long as the entry is below the 11th rib, the advantages generally outweigh the risks. Percutaneous access into an infundibulum or the renal pelvis poses a greater risk of vascular injury than calyceal entry. Middle calyceal access provides good access to the ureter but usually does not provide good access to the upper and lower calyces. In the prone position an anterior calyx offers little access to the rest of the kidney.

11. **e. Retrograde placement of externalized (single pigtail) ureteral stent for drainage.** This can be performed at the conclusion of the procedure for drainage as an alternative to a nephrostomy tube, but it is not useful before access because the pigtail might interfere with the procedure and it would not have any advantage over a straight ureteral catheter or an occlusion balloon catheter. The other choices are all well-described techniques of retrograde assistance for percutaneous renal access.
12. **e. Is more traumatic.** A 21-gauge needle is not as easy to direct as an 18-gauge needle because it is more flexible. A 21-gauge needle requires a 0.018-inch guidewire, and because of this extra step (exchanging the 0.018-inch guidewire for a 0.035-inch guidewire) there is a greater risk of loss of access. Compared with an 18-gauge needle, the 21-gauge needle is less traumatic; this is its primary advantage, and it is for this reason that the 21-gauge needle should be used when the operator is less experienced or if minimizing trauma is paramount.
13. **d. Visualizes the access needle better.** It is easier to see a needle and monitor tract dilation with fluoroscopy than with ultrasonography. Percutaneous access is always more difficult in the morbidly obese, and ultrasonography is no better than fluoroscopy in this situation. Ultrasonography is more portable, can more rapidly evaluate different views of the kidney, and is preferred in settings in which retrograde access cannot be attained or is difficult to attain (kidneys above urinary diversions, transplanted kidneys, kidneys above a completely obstructed ureter, etc.).
14. **e. Continuously monitors depth of needle penetration.** The "triangulation" technique monitors depth of needle placement in all fluoroscopic views, whereas the "eye-of-the-needle" technique assesses depth only at the final step. If the fluoroscopy field is collimated down and the needle is held with a hemostat, sponge forceps, or purpose-built needle holder, then radiation exposure to the operator's hands can be avoided with both techniques. Retrograde assistance is useful with any fluoroscopic percutaneous

renal access, and both techniques are more difficult in morbidly obese patients.

15. **c. Least expensive with metal dilators.** Metal dilators are least expensive on a per-case basis because they are reusable. The metal dilators are also the most effective dilators. It is uncertain which are the safest dilators, at least in terms of association with hemorrhage. The balloon dilators are more effective in hypermobile kidneys than other techniques and also more rapid than sequential passage of metal or semirigid dilators. The one-shot semirigid dilator technique requires considerable manual force to create the tract.
16. **e. Lower hemorrhage rates than in normal kidneys can be expected.** The rate of major hemorrhagic complications during percutaneous renal surgery in horseshoe kidneys (4.3%) is less than in normal kidneys (6% to 20%). Upper pole access is useful, and direct lower pole access is usually not possible. Cross-sectional imaging is useful in assessing the anatomy of horseshoe kidneys. **The initial entry into a horseshoe kidney usually is more medial than in normal kidneys.**
17. **a. Retrograde assistance is difficult.** Because of the site of ureteral implantation in the bladder, retrograde assistance for percutaneous renal surgery in transplant kidneys is difficult. Given this, ultrasonography is more useful than fluoroscopy for guiding initial access. Semirigid plastic and metal dilators are often more useful than balloon dilators because of the perinephric scarring, which makes the transplant kidney quite fixed in place rather than hypermobile. Despite the challenges, percutaneous renal surgery in transplant kidneys has a high success rate, and secondary procedures are usually not required.
18. **d. Stay more securely in the kidney than Malecot catheters.** Malecot tubes are the easiest to pull out, and circle nephrostomy tubes are the hardest. All tubes should be secured at the skin to reduce the risk of at least one mechanism of tube removal. A ureteral catheter can be passed through the end hole of a Councill catheter; a Foley catheter does not have this end hole. Saline or water should be used to inflate the balloon, because the more

viscous contrast material might hinder emptying of the balloon when removal is attempted. All nephrostomy tubes, even ones with robust internal retention devices, should be fixed to the skin externally with a suture or other mechanism. There is no evidence that Foley and Malecot catheters differ in propensity for infection.

19. **a. Is used in nephroureteral stents.** The Cope retention mechanism is used in the renal pelvic portion of nephroureteral stents. According to one study, the Cope retention mechanism is more secure than Malecot wings but does not retain as well as a balloon. It is more secure than a passive pigtail retention mechanism owing to the string that holds the coil in place. There is no evidence that more than one Cope-type nephrostomy tube should not be used.
20. **a. Maintenance of the working sheath.** The stiff working access sheath would be a poor choice for postprocedure collecting system drainage. All of the other options have been described.
21. **a. Offers greater assurance of upper urinary tract drainage than an internal ureteral stent.** Drainage of upper urinary tract after percutaneous renal surgery is adequate with an internal ureteral stent in most cases (or with no tube at all in selected cases), but when hemorrhage occurs, the larger caliber of a nephrostomy tube provides better drainage of the upper urinary tract collecting system than an internal ureteral stent. The nephrostomy tube does not have to be placed in the dilated access site (i.e., it can be placed at a new site), although that is common practice. Although redilation may be required, any external nephrostomy tube maintains the percutaneous access tract. There is actually less hemorrhage when a postoperative nephrostomy tube is omitted. Most studies suggest that the pain associated with nephrostomy tubes is related to tube diameter, with smaller-caliber tubes causing less pain.
22. **d. Less need for removal in the radiology suite.** The removal of larger tubes occasionally can be followed by immediate hemorrhage; this is rare with smaller tubes. Therefore large-caliber nephrostomy tubes should be removed in a radiology suite where there is the opportunity for immediate replacement of the tube.

Small-caliber tubes can be removed safely at the bedside after a period of clamping to assess clinically for distal ureteral obstruction. A number of studies have compared the impact of nephrostomy tube diameter after percutaneous renal surgery. Only one study found no benefit to the smaller tube. Otherwise, consistent advantages of the small-caliber tubes were less pain, less urinary leakage, and no change in postprocedure blood loss. There is no consistent evidence that small-caliber tubes are associated with shorter duration of hospitalization compared with large-caliber tubes.

23. **b. Microwave treatment of the tract.** Microwave treatment of the tract would be difficult with current instruments. The other options have all been described. Other hemostatic agents that have been inserted/instilled into the tract include gelatin sponge and gelatin granules plus thrombin.
24. **c. Greater narcotic use.** Most randomized controlled trials comparing internal ureteral stents to large-caliber nephrostomy tubes after percutaneous renal surgery have shown reduced narcotic use in the stented patients. The difference is less significant when a small-caliber nephrostomy tube is used. Depending on physician preference, both internal ureteral stents (if attached to a string that exits via the flank) and small-caliber nephrostomy tubes can be removed at the bedside. Randomized controlled trials comparing internal ureteral stents to nephrostomy tubes have not revealed any difference in technical success rates, complication rates, or incidence of urinary leakage from the skin entry site.
25. **d. Occlude the nephrostomy tube and apply pressure to the incision.** The first step in this situation is to occlude the nephrostomy tube and apply pressure to the incision. Let the collecting system clot off, and do not irrigate until the following morning. This management is successful in the majority of cases. If bleeding persists, then insert a Kaye nephrostomy tamponade balloon. An 18-Fr Malecot catheter will be no more effective than the 12-Fr Cope nephrostomy tube. Irrigation is not useful, and removing the nephrostomy tube altogether is ill advised.

26. **a. Continue with the procedure if vision is adequate.** If the procedure can be continued with acceptable vision, then the blood loss cannot be great. However, if vision is lost, then the procedure must be aborted. If so, then inserting and occluding a nephrostomy tube, as well as applying pressure to the incision so that the collecting system clots off, will suffice in most cases. If this is not successful, then place a Councill catheter and attempt to inflate the balloon at the injury site. Instillation of gelatin granules plus thrombin into the collecting system can create a clot that is difficult to manage. Selective angioembolization is required only when an arterial injury does not respond to less intensive management, or if the injury is obviously a significant one that will not respond to these maneuvers.
27. **e. Come to the hospital.** Any report of bright red blood in the urine after percutaneous renal surgery should prompt hospital admission. This man likely has an arteriovenous fistula or arterial pseudoaneurysm. The conservative measures are not likely to be helpful, and aminocaproic acid (Amicar) is contraindicated in the setting of upper tract hemorrhage.
28. **c. Massive hemorrhage.** The renal pelvis will clot off before the pressure from hemorrhage would rupture it. Any manipulation during percutaneous renal surgery can cause renal pelvic perforation.
29. **d. Back out the nephrostomy tube into the colon and insert a new nephrostomy tube. The main principle of care of a colon injury associated with percutaneous renal surgery is prompt and separate drainage of the colon and urinary collecting system.** If detected postoperatively, the simplest management is to back the nephrostomy tube out of the kidney and into the colon to serve as a colostomy, and then obtain separate access to the upper urinary tract, with either a new percutaneous access that does not traverse the colon or a retrograde-placed ureteral stent. Parenteral feeding is usually not required, and for the typical extraperitoneal injury, open surgical repair usually is needed only if the patient develops peritonitis or sepsis.

30. **a. Liver.** Although splenic injuries have been managed conservatively, the need for surgical intervention is more likely than with liver injuries. Injuries to small bowel or the biliary system require prompt treatment.
31. **d. Combined with distal ureteral obstruction, a nephropleural fistula can occur.** Nephropleural fistula (urinothorax) is a direct and persistent communication between the intrarenal collecting system and the intrathoracic cavity, which can follow percutaneous renal access of the upper urinary tract in the setting of pleural transgression. Some degree of distal ureteral obstruction usually contributes to the problem. The rates of pleural injures for infra-12th rib, supra-12th rib, and supra-11th rib punctures are approximately less than 0.5%, 5%, and 25%, respectively. Thoracostomy is not necessary for all patients with hydrothorax. If one is needed, then a small-caliber tube with a Heimlich valve is all that is required in the absence of lung injury.
32. **c. Can have fatal consequences. Intravascular hemolysis from the extravasated water irrigant can be fatal. The irrigant for percutaneous renal surgery should be saline, with the exception of glycine or similar nonelectrolytic isotonic fluids when monopolar electrocautery is used.** Intravascular or extravascular extravasation of fluid from continued irrigation in the setting of a large venous injury or collecting system perforation can lead to clinically significant sequelae including volume overload and extrarenal fluid collections that require drainage.
33. **a. Observation.** Preoperative and perioperative management of this patient has been appropriate. In this setting, most patients with fever after percutaneous nephrolithotomy do not have infection. If the fever is an isolated postoperative one, then standard postoperative care (early ambulation, use of incentive spirometry, etc.) is all that is needed. If the fever does not resolve promptly, then appropriate diagnostic evaluation and initiation of antimicrobial therapy and other supportive care are indicated.
34. **b. Minimal in the absence of vascular injury.** The kidney suffers little permanent damage after uncomplicated percutaneous renal surgery. If there is significant loss of function, it is usually owing to

disastrous vascular injury or the angioembolization used to treat hemorrhage. Loss of renal function associated with percutaneous renal surgery is less than or equal to the loss associated with shock wave lithotripsy. There is no evidence that damage to the kidney is any more or less in pelvic, orthotopic, solitary, or nonsolitary kidneys.

35. **e. None of the above.** Randomized controlled trials have failed to show association between stent length, diameter, material, or architecture and impact on patient tolerability.

Chapter Review

1. Percutaneous nephrostomy and retrograde ureteral stents are generally equivalent in their capacity to resolve fever in patients with upper urinary tract obstruction. However, obstruction complicated by infection is an emergency, and in the unstable patient, percutaneous drainage may be more efficacious.
2. The colon can be lateral or posterior to the right and left kidney.
3. A guidewire that enters the kidney percutaneously and exits the urethra via the meatus (through and through access) may be the only guidewire used when operating on the upper urinary tract. However, in all other situations, two guidewires—a safety and a working guidewire—are required. No matter what the access, it is always prudent to have a safety guidewire in addition to the working guidewire.
4. It is imperative that the dilators do not pass too far into the collecting system because this results in renal pelvic injury.
5. Percutaneous nephrostomy is generally the preferred approach for endoscopy of the obstructed collecting system in the transplanted kidney.
6. Approximately 1% of percutaneous procedures are complicated by delayed hemorrhage. Delayed hemorrhage is usually due to arteriovenous fistulas or arterial pseudoaneurysms. The preferred management is selective angioembolization.

7. Renal arteries are end arteries and result in loss of the segment of renal parenchyma they supply when occluded. Renal veins communicate with each other.
8. Complications of percutaneous nephrostomy include hemorrhage, collecting system injury, colon injury, pleural injury, neuromuscular injuries, air embolism, and infundibular stenosis.
9. Although compound calyces are common in the lower pole, they are almost always present in the upper pole.
10. The recommended preoperative cessation periods are as follows: herbal medicines, 1 week; clopidogrel, 5 days; aspirin, 1 week; warfarin, 5 days; nonsteroidal inflammatory agents, 3 to 7 days.
11. The main principle of care of a colon injury associated with percutaneous renal surgery is prompt and separate drainage of the colon and urinary collecting system. The simplest management is to back the nephrostomy tube out of the kidney and into the colon to serve as a colostomy, and then obtain separate access to the upper urinary tract, either with a new percutaneous access that does not traverse the colon or with a retrograde-placed ureteral stent.
12. Intravascular hemolysis from the extravasated water irrigant can be fatal. The irrigant for percutaneous renal surgery should be saline, with the exception of glycine or similar nonelectrolytic isotonic fluids when monopolar electrocautery is used.